

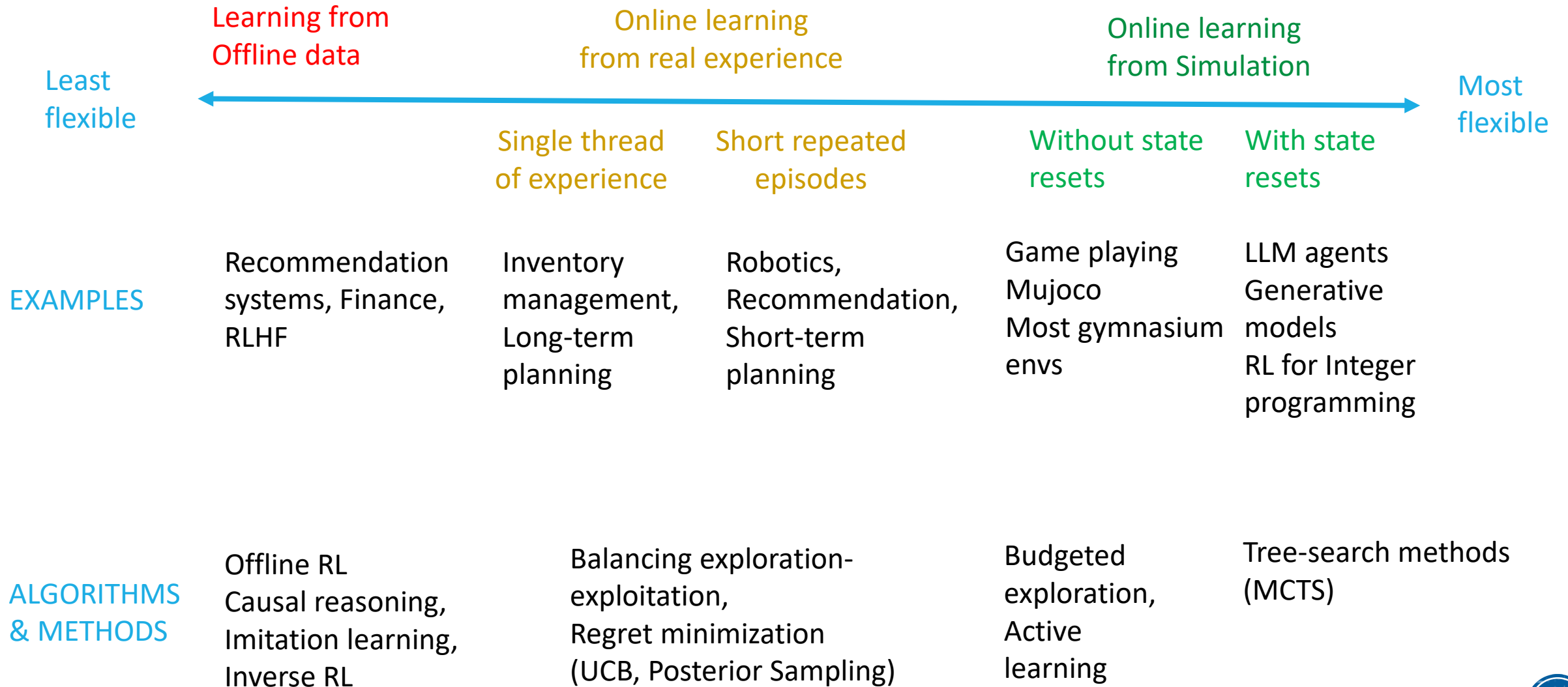
Exploration-exploitation and theoretical guarantees in Reinforcement Learning

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ORCS 4529 (Reinforcement Learning)



RL environments and algorithms



Quantifying the performance of RL algorithms

- Computational complexity
 - the amount of per-time-step computation the algorithm uses;
- Space complexity
 - the amount of memory used by the algorithm;
- Learning complexity
 - How much samples/information does the algorithm need to collect?
 - Can it learn effectively from feedback generated by reasonable actions?

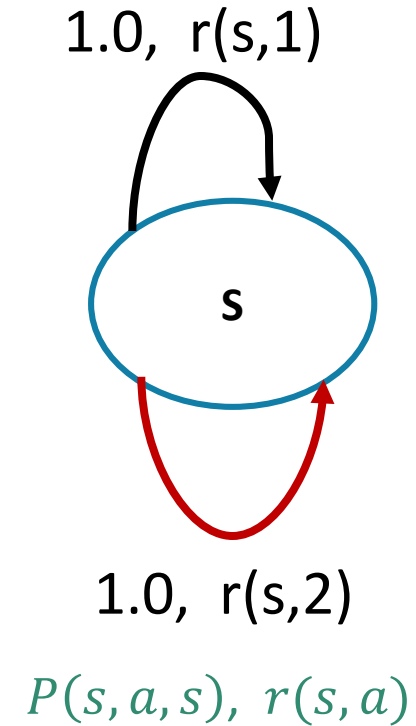
Understanding learning complexity of an RL algorithm

What is the optimal learning complexity for a given (distribution over) problem instance(s)?

Is there an optimal learning algorithm?

Multi-armed bandit == single state MDP

Two armed bandit



Stochastic multi-armed bandit problem

- Online decisions
 - At every time step $t = 1, \dots, T$, pull one arm out of N arms
- Stochastic feedback
 - For each arm i , reward is generated i.i.d. from a **fixed but unknown distribution** ν_i support $[0,1]$, mean μ_i
- Bandit feedback
 - Only the reward of the pulled arm can be observed
- Optimal solution: always play arm with highest μ_i (apriori unknown)
 - Need to learn $\mu_i, i = 1, \dots, n$ to find $\arg \max \mu_i$



The multi-armed bandit problem (Thompson 1933; Robbins 1952)

Multiple rigged slot machines in a casino.

Which one to put money on?

- Try each one out

Arm == actions



***WHEN TO STOP TRYING (EXPLORATION) AND START PLAYING
(EXPLOITATION)?***



Optimal learning

- Learning complexity depends on the instance
 - It is easy to learn if all the machines are really bad compared to one
- Given a prior distribution over instances, find an algorithm to
 - (implicitly explicitly) resolve what kind of instance you are in
 - Find the optimal arm for that instance

What is the best way to achieve this?

Gittins Index

It's a planning problem:

- draw a tree of actions
- how much information you will have at every stage and what rewards you would have achieved
 - For every arm i : number of plays n_i , number of successes α_i
- Seminal paper by Gittins provides an index based policy to find paths in this tree that maximize expected discounted reward $E[\sum_{t=1}^{\infty} \gamma^{t-1} r_t]$
- Tractable in special cases ($\gamma < 1$, *MAB*, Gittins, 1989).

Approximate learning measures

- Instead of maximizing expected reward provide bounds on regret
- Regret = Algorithm's total reward compared to running the optimal policy from the beginning
- MAB: expected regret in time T
 - $Regret(T) = T\mu^* - E[\sum_{t=1}^T r_t]$ where $\mu^* = \max_j \mu_j$
- Regret is number of mistakes. Can we achieve regret sublinear in time T ? logarithmic?

Other measures

- Sample complexity
- PAC (Probably bounds

Number of samples required to achieve an ϵ -optimal policy with probability $1 - \delta$ at the end of learning phase.

Learning separate from execution (simulation environments)

Theoretically just smart exploration -- but exploration-exploitation techniques still effective due to need to for focusing in on high reward policies

Overview

- Exploration-exploitation and regret minimization for multi-armed bandit
 - UCB (Upper Confidence Bounds)
 - Thompson Sampling
- Regret minimization for RL
 - UCRL (Upper Confidence RL)
 - Posterior sampling
 - Optimistic Q-learning
- Other techniques
 - MCTS
 - Offline RL

Recall: Stochastic multi-armed bandit problem

- Online decisions
 - At every time step $t = 1, \dots, T$, pull one arm out of N arms
- Stochastic feedback
 - For each arm i , reward is generated i.i.d. from a **fixed but unknown distribution** ν_i support $[0,1]$, mean μ_i
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Regret minimization problem in the multi-armed bandit (MAB)

- Minimize expected regret in time T
 - $Regret(T) = T\mu^* - E[\sum_{t=1}^T r_t]$ where $\mu^* = \max_j \mu_j$
- Equivalent formulation
 - Expected regret for playing arm $I_t = i$ at time t : $\Delta_i = \mu^* - \mu_i$
 - $n_{i,T}$ be the number times arm i is played till time T
 - $Regret(T) = \sum_t (\mu^* - \mu_{I_t}) = \sum_{i: \mu_j \neq \mu^*} \Delta_i E[n_{i,T}]$



What can we expect?

- If we can bound $n_{i,T}$ by $\frac{C \log(T)}{\Delta_i^2}$, then regret bound: $\sum_{i:\mu_i \neq \mu^*} \frac{C \log(T)}{\Delta_i}$
 - Problem-dependent bound, close to lower bound [Lai and Robbins 1985]
- Problem independent bound: $C\sqrt{NT\log(T)}$
 - Separately bound total regret for playing an arm with $\Delta_i \leq \sqrt{\frac{N\log(T)}{T}}$ and $\Delta_i >$
 $\sqrt{\frac{N\log(T)}{T}}$
 - Lower bound $\Omega(\sqrt{NT})$,

The need for exploration

- Two arms **black** and **red**
 - Random rewards with unknown mean $\mu_1 = 1.1$, $\mu_2 = 1$
 - Optimal expected reward in T time steps is $1.1 \times T$
- Exploit only strategy: use the current best estimate (MLE/empirical mean) of unknown mean to pick arms
- Initial few trials can mislead into playing red action forever

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- Expected regret in T steps is close to $0.1 \times T$

Exploration-Exploitation tradeoff

- Exploitation: play the empirical mean reward maximizer
- Exploration: play less explored actions to ensure empirical estimates converge

Upper confidence bound algorithms for MAB [Auer 2002]

Optimism under face of uncertainty

- Empirical mean at time t for arm i

$$\hat{\mu}_{i,t} = \frac{\sum_{s=1: I_s=i}^t r_s}{n_{i,t}}$$

- Upper confidence bound

$$\text{UCB}_{i,t} := \hat{\mu}_{i,t} + 2\sqrt{\frac{\ln t}{n_{i,t}}}$$

Algorithm 1: UCB algorithm for the stochastic N-armed bandit problem

```
foreach  $t = 1, \dots, N$  do
  Play arm  $t$ 
end
foreach  $t = N + 1, N + 2, \dots, T$  do
  Play arm  $I_t = \arg \max_{i \in \{1 \dots N\}} \text{UCB}_{i,t-1}$ .
  Observe  $r_t$ , compute  $\text{UCB}_{i,t}$ 
end
```

Regret analysis

- Using Azuma-Hoeffding (since $E[r_t | I_t = i] = \mu_i$), with probability $1 - \frac{2}{t^2}$,

$$|\hat{\mu}_{i,t} - \mu_i| < \sqrt{\frac{4 \ln t}{n_{i,t}}}$$

- Two implications: for every arm i

- $UCB_{i,t} > \mu_i$ with probability $1 - \frac{2}{t^2}$

Recall: $UCB_{i,t} := \hat{\mu}_{i,t} + 2\sqrt{\frac{\ln t}{n_{i,t}}}$

- If $n_{i,t} > \frac{16 \ln(T)}{\Delta_i^2}$, $\hat{\mu}_{i,t} < \mu_i + \frac{\Delta_i}{2}$, $UCB_{i,t} < \hat{\mu}_{i,t} + \frac{\Delta_i}{2}$

- Each suboptimal arm is played at most $\frac{16 \ln(T)}{\Delta_i^2}$ times, since after that many plays:

$$\begin{aligned} UCB_{i,t} &= \hat{\mu}_{i,t} + \sqrt{\frac{\ln t}{n_{i,t}}} \leq \hat{\mu}_{i,t} + \frac{\Delta_i}{2} \\ &< \left(\mu_i + \frac{\Delta_i}{2} \right) + \frac{\Delta_i}{2} \\ &= \mu^* \\ &< UCB_{i^*,t} \end{aligned}$$

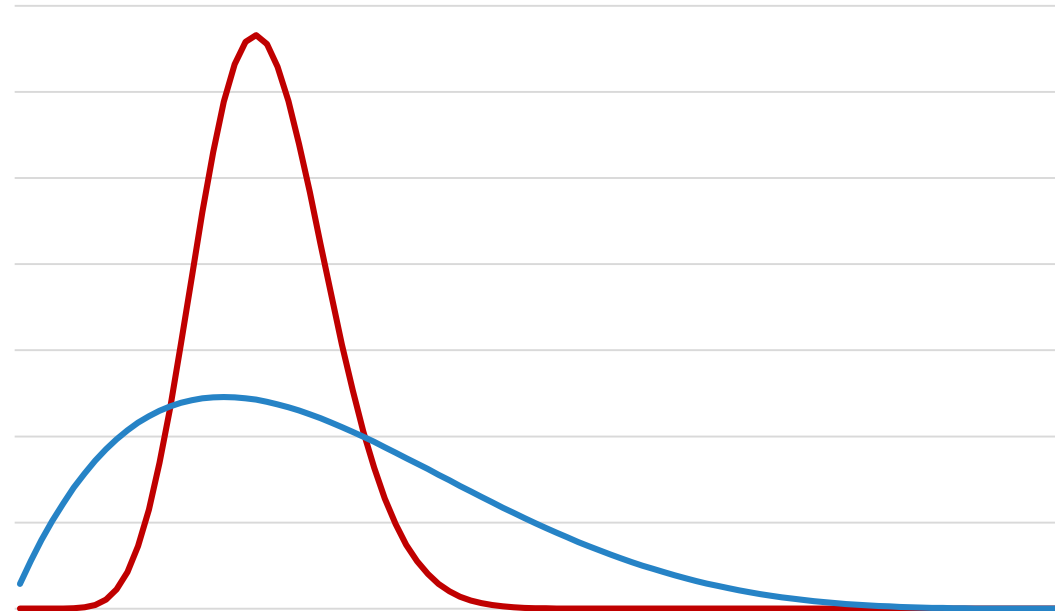
Thompson Sampling [Thompson, 1933]

- Natural and Efficient heuristic
- Maintain belief about parameters (e.g., mean reward) of each arm
- Observe feedback, update belief of pulled arm i in Bayesian manner
- Pull arm with posterior probability of being best arm
 - NOT same as choosing the arm that is most likely to be best



Posterior Sampling: main idea [Thompson 1933]

- Maintain Bayesian posteriors for unknown parameters
- With more trials posteriors concentrate on the true parameters
 - Mode captures MLE: enables exploitation
- Less trials means more uncertainty in estimates
 - Spread/variance captures uncertainty: enables exploration
- A sample from the posterior is used as an estimate for unknown parameters to make decisions

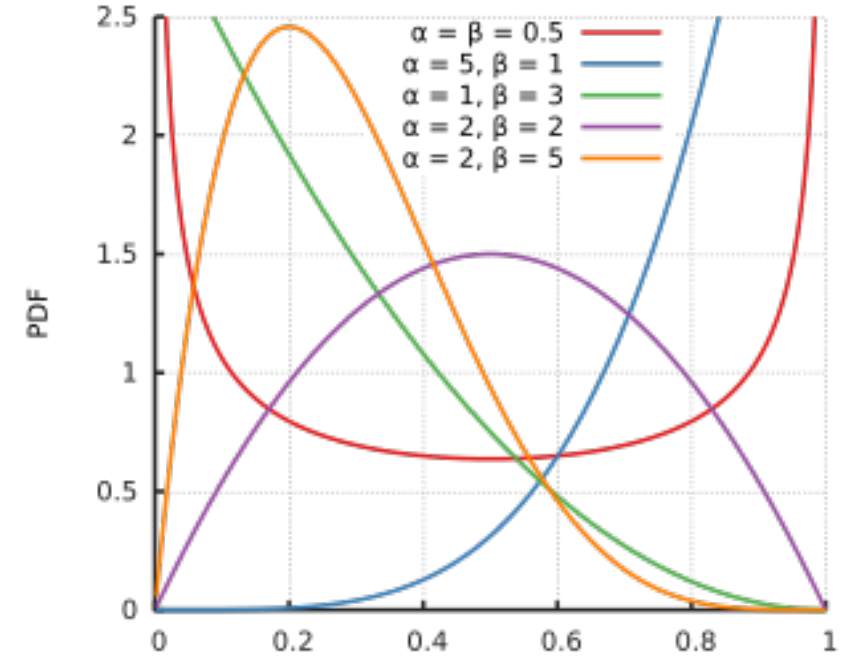


Bernoulli rewards, Beta priors

Uniform distribution $Beta(1,1)$

$Beta(\alpha, \beta)$ prior $\Rightarrow$ Posterior

- $Beta(\alpha + 1, \beta)$ if you observe 1
- $Beta(\alpha, \beta + 1)$ if you observe 0



Start with $Beta(\alpha_0, \beta_0)$ prior belief for every arm

In round t ,

- For every arm i , sample $\theta_{i,t}$ independently from posterior $Beta(\alpha_0 + S_{i,t}, \beta_0 + F_{i,t})$
- Play arm $i_t = \max_i \theta_{i,t}$
- Observe reward and update the Beta posterior for arm i_t



Bayesian regret bounds for Thompson Sampling

- Theorem [Russo and Van Roy 2014]: Given any prior f over MAB instance M described by reward distributions with $[0,1]$ bounded support (e.g., prior distribution over $\mu_1, \mu_2, \dots, \mu_N$ in case of Bernoulli rewards)

$$\text{BayesianRegret}(T) = E_{M \sim f}[\text{Regret}(T, M)] \leq O(\sqrt{NT \ln T})$$

Note;

- $\text{Regret}(T, M)$ notation is used to explicitly indicate dependence on regret on the MAB instance M .
- In comparison, worst case regret,

$$\text{Regret}(T) = \max_M [\text{Regret}(T, M)]$$

Worst-case regret bounds

[A. and Goyal COLT 2012, AISTATS 2013, JACM 2017, Kaufmann et al. 2013]

Instance-dependent bounds for $\{0,1\}$ rewards

- $\text{Regret}(T) \leq \ln(T)(1 + \epsilon) \sum_i \frac{\Delta_i}{KL(\mu^* || \mu_i)} + O\left(\frac{N}{\epsilon^2}\right)$
 - Matches *asymptotic instance wise lower bound* [Lai Robbins 1985]
 - UCB algorithm achieves this only after careful tuning [Kaufmann et al. 2012]

Arbitrary bounded $[0,1]$ rewards (using Beta and Gaussian priors)

- $\text{Regret}(T) \leq O(\ln(T) \sum_i \frac{1}{\Delta_i})$
 - Matches the best available for UCB for general reward distributions

Instance-independent bounds (Beta and Gaussian priors)

- $\text{Regret}(T) \leq O(\sqrt{NT \ln T})$
- Prior and likelihood mismatch allowed!



Why does it work? Two arms example

- Two arms, $\mu_1 \geq \mu_2$, $\Delta = \mu_1 - \mu_2$
- Every time arm 2 is pulled, Δ regret
- ➡ ▪ Bound the number of pulls of arm 2 by $\frac{\log(T)}{\Delta^2}$ to get $\frac{\log(T)}{\Delta}$ regret bound
- How many pulls of arm 2 are actually needed?



Easy situation

After $n = O(\frac{\log(T)}{\Delta^2})$ pulls of arm 2 **and arm 1**

- Empirical means are well separated

$$\text{Error } |\widehat{\mu}_i - \mu_i| \leq \sqrt{\frac{\log(T)}{n}} \leq \frac{\Delta}{4} \text{ whp}$$

(Using Azuma Hoeffding inequality)

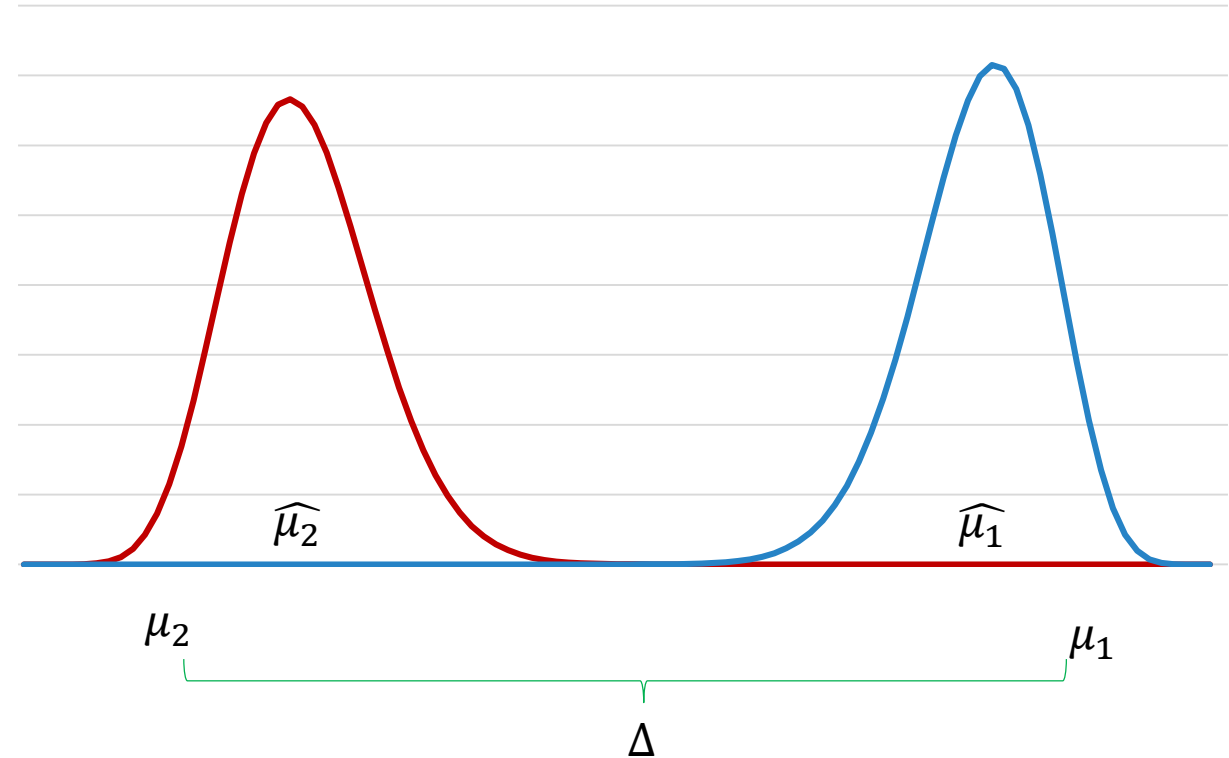
- Beta Posteriors are well separated

$$\text{Mean} = \frac{\alpha_i}{\alpha_i + \beta_i} = \widehat{\mu}_i$$

$$\text{standard deviation} \simeq \frac{1}{\sqrt{\alpha + \beta}} = \frac{1}{\sqrt{n}} \leq \frac{\Delta}{4}$$

The two arms can be distinguished!

No more arm 2 pulls.



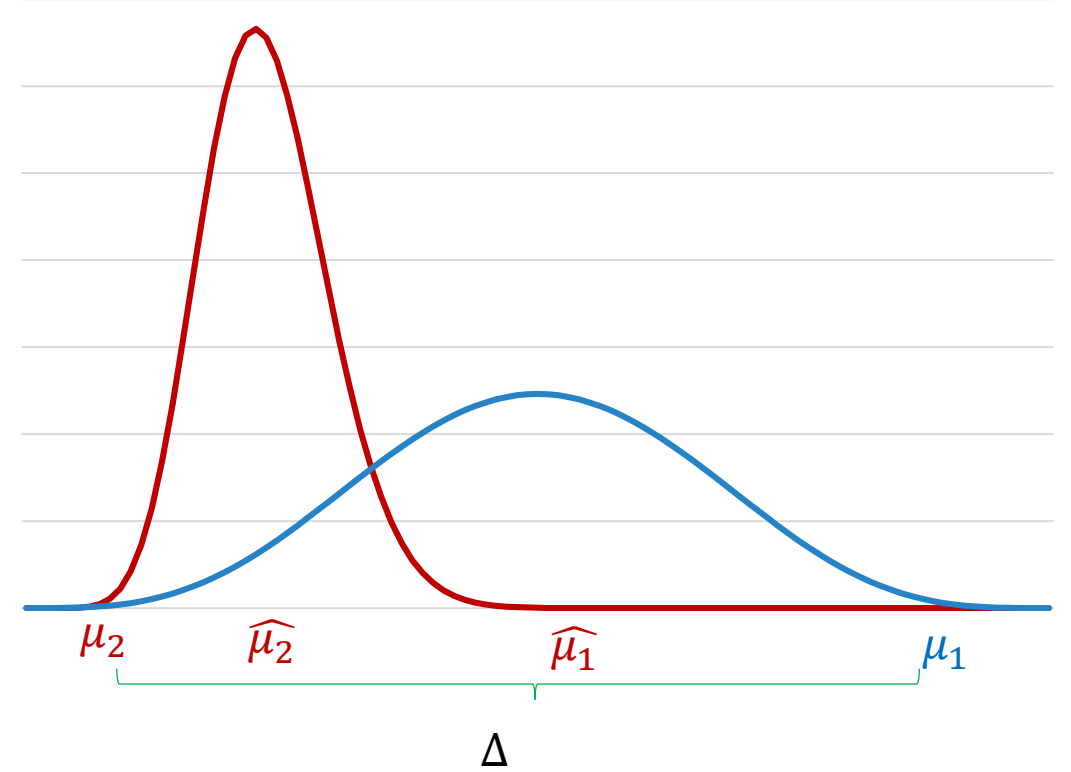
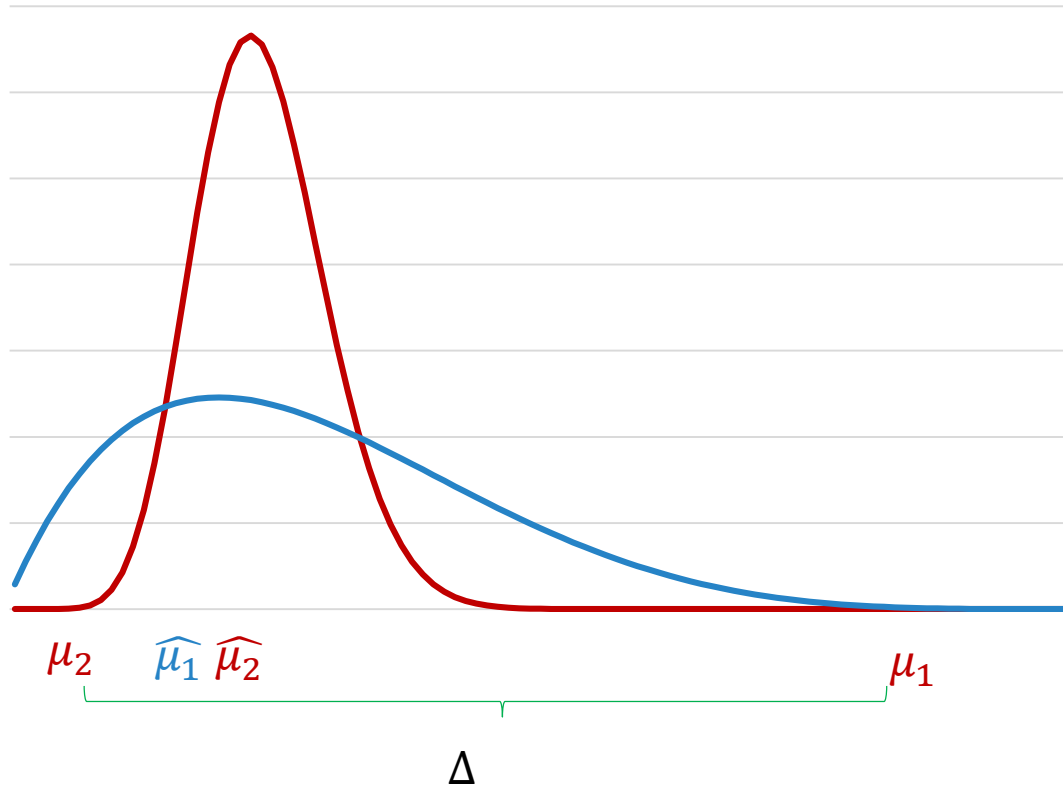
Easy situation

- If arm 2 is pulled less than $n = O\left(\frac{\log(T)}{\Delta^2}\right)$ times?
 - Regret is at most $n\Delta = \frac{\log(T)}{\Delta}$



Difficult situation

- At least $\frac{\log(T)}{\Delta^2}$ pulls of arm 2, but few pulls of arm 1



Main insight

- Arm 1 will be played roughly every constant number of steps in this situation
- It will take at most $constant \times \frac{\log T}{\Delta^2}$ steps (extra pulls of arm 2) to get out of this situation
- Total number of pulls of arm 2 is at most $O(\frac{\log T}{\Delta^2})$
- Summary: variance of posterior enables exploration
- Optimal bounds (and for multiple arms) require more careful use of posterior structure



Next...

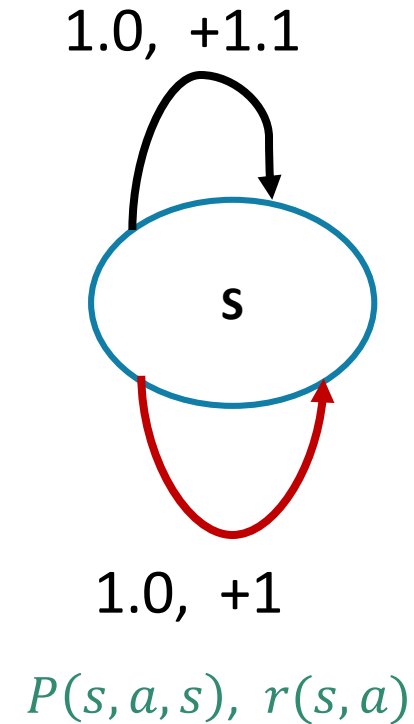
Using UCB and TS ideas for exploration-exploitation in RL

The need for exploration

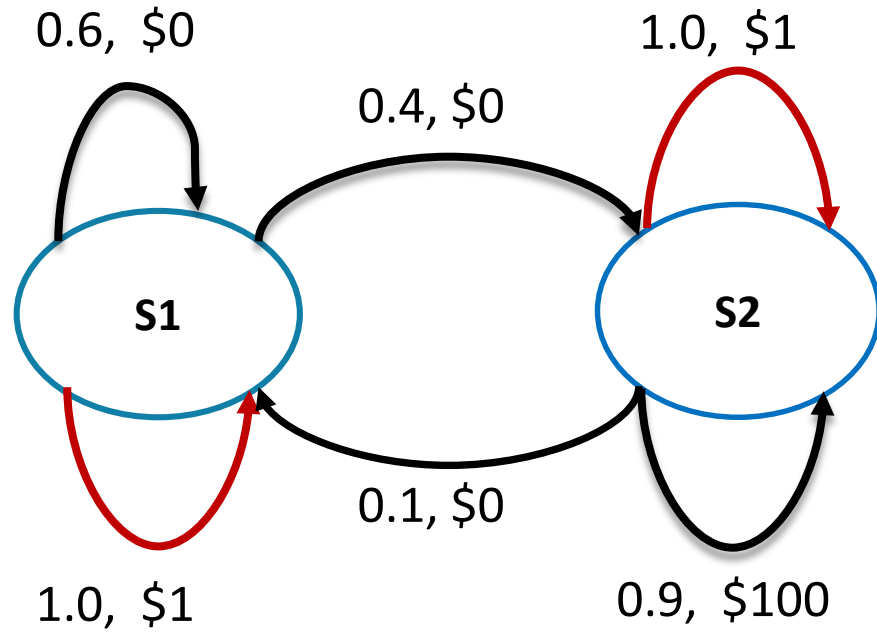
- Single state MDP
 - Solution concept: optimal action
 - **Multi-armed bandit problem**
- Uncertainty in rewards
 - Random rewards with unknown mean μ_1, μ_2
- Exploit only: use the current best estimate (MLE/empirical mean) of unknown mean to pick arms
- Initial few trials can mislead into playing red action forever

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The need for exploration



- Uncertainty in rewards, state transitions
- Unknown reward distribution, transition probabilities
- Exploration-exploitation:
 - Explore actions/states/policies, learn reward distributions and transition model
 - Exploit the (seemingly) best policy

Summary of recent work: episodic setting

T is total number of steps, H steps per episode

Optimistic Value iteration [Azar, Osband, Munos, 2017]

- Model based algorithm, Worst-case regret bound $\tilde{O}(H\sqrt{SAT})$

Optimistic Q-learning [Jin et al. 2018]

- Model-free, Worst-case regret bound $\tilde{O}(\sqrt{H^3SAT})$

Posterior sampling known prior setting [Osband, Russo, and Van Roy 2013, Osband and Van Roy, 2016, 2017]

- Model-based algorithm, **Bayesian regret** bound of $\tilde{O}(H\sqrt{SAT})$

(Optimistic) Posterior sampling unknown prior [Tiapkin et al. 2022]

- Model-based algorithm, Worst-case regret bound $\tilde{O}(H^3\sqrt{SAT})$

Summary of recent work: non-episodic setting

Communicating MDP diameter D

Upper confidence bound based algorithms [Jaksch, Ortner, Auer, 2010]
[Bartlett, Tewari, 2012]

- Worst-case regret bound $\tilde{O}(DS\sqrt{AT})$
- Lower bound $\Omega(\sqrt{DSAT})$

Optimistic Posterior Sampling [A. Jia 2017]

- Worst-case regret bound $\tilde{O}(DS\sqrt{AT})$

Both of the above are model-based algorithms

Next...

- **UCRL: Upper confidence bound based algorithm for RL**
- Posterior sampling based algorithm for RL
 - Main result
 - Proof techniques

UCRL algorithm [Jacksch, Ortner, Auer 2002]

- Similar principles as UCB

This is a ***Model-based approach***

- Maintain an estimate of model $\hat{P}, \hat{R}$
- Occasionally solve the MDP $(S, A, \hat{P}, \hat{R}, s_1)$ to find a policy
- Run this policy for some time to get samples, and update model estimate

Compare to “model-free” approach or direct learning approach like Q-learning

- Directly update Q-values or value function or policy using samples.

UCRL algorithm

- Proceed in epochs, an epoch ends when the number of visits of *some* state-action pair doubles.

In the beginning of every epoch k

- Use samples to compute an optimistic MDP $(S, A, \tilde{R}, \tilde{P}, s_1)$
 - MDP with value greater than true MDP (Upper bound!!)
- Solve the optimistic MDP to find optimal policy $\tilde{\pi}$

Execute $\tilde{\pi}$ in epoch k

- observe samples s_t, a_t, r_t, s_{t+1}

Go to next epoch If visits of *some* state-action pair doubles

- If $n_k(s, a) \geq 2 n_{k-1}(s, a)$ for some s, a

UCRL algorithm (computing optimistic MDP)

In the beginning of every epoch k

- For every s, a , compute **empirical** model estimate
 - let $n_k(s, a)$ be the number of times s, a was visited before this epoch,
 - let $n_k(s, a, s')$ be the number of transition to s'
 - Set $\hat{R}(s, a)$ as the empirical average rewards over these $n_k(s, a)$ steps
 - Set $\hat{P}(s, a, s')$ as $\frac{n_k(s, a, s')}{n_k(s, a)}$
- Compute **optimistic** model estimate
 - Use Chernoff bounds to define confidence region around $\hat{R}, \hat{P}$
 - $|\hat{R}(s, a, s') - R(s, a, s')| \leq \frac{R_{max} \log(t)}{\sqrt{n_k(s, a)}}$ with probability $1 - \frac{1}{t^2}$
 - $|\hat{P}(s, a, s') - P(s, a, s')| \leq \frac{\log(t)}{\sqrt{n_k(s, a)}}$ with probability $1 - \frac{1}{t^2}$
 - True R, P lies in this region
 - Find the best combination, $\tilde{R}, \tilde{P}$ in this region
 - MDP $(S, A, \tilde{R}, \tilde{P}, s_1)$ with maximum value
 - Will have value more than the true MDP

Main result

- Regret:

$$\text{Regret}(M, T) := T \rho^* - \sum_{t=1}^T r(s_t, a_t)$$

- Theorem: For any **communicating** MDP M with (unknown) diameter D , with high probability:

$$\text{Regret}(M, T) \leq \tilde{O}(DS\sqrt{AT})$$

- $\tilde{O}(\cdot)$ notation hides logarithmic factors in S, A, T beyond constants.

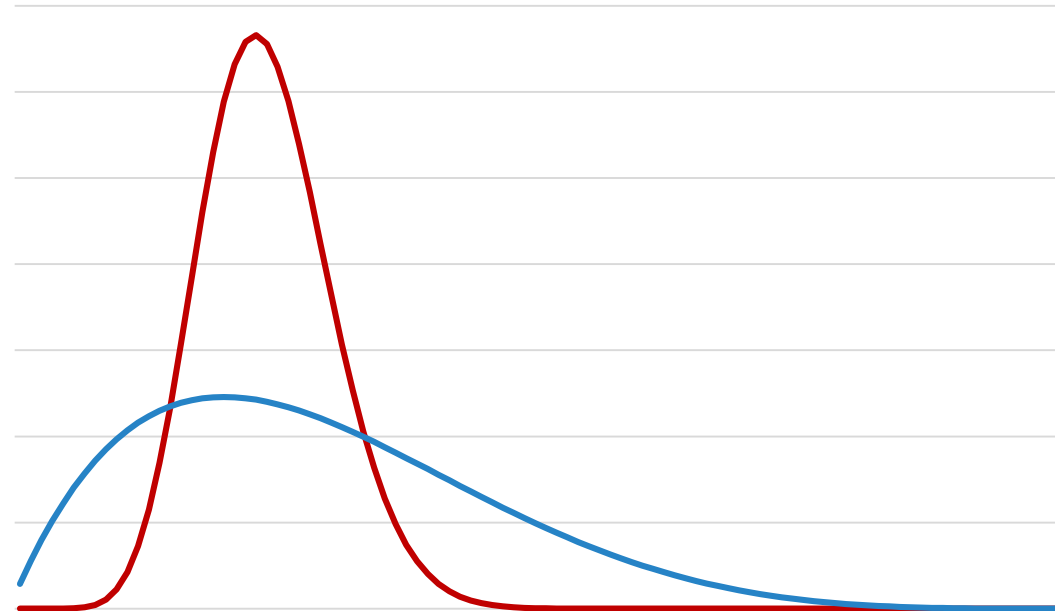
Next...

➤ **Posterior sampling algorithm** for MDPs

- Main result
- Proof techniques

Posterior Sampling: main idea [Thompson 1933]

- Maintain Bayesian posteriors for unknown parameters
- With more trials posteriors concentrate on the true parameters
 - Mode captures MLE: enables exploitation
- Less trials means more uncertainty in estimates
 - Spread/variance captures uncertainty: enables exploration
- A sample from the posterior is used as an estimate for unknown parameters to make decisions



Posterior Sampling : Bayesian posteriors

- Assume for simplicity: Known reward distribution
- Needs to learn the unknown transition probability vector $P_{s,a} = (P_{s,a}(1), \dots, P_{s,a}(S))$ for all s, a
- In any state $s_t = s, a_t = a$, observes new state s_{t+1}
 - outcome of a Multivariate Bernoulli trial with probability vector $P_{s,a}$

Posterior Sampling with Dirichlet priors

- Given prior $\text{Dirichlet}(\alpha_1, \alpha_2, \dots, \alpha_S)$ on $P_{s,a}$
- After a Multinoulli trial with outcome (new state) i , Bayesian posterior on $P_{s,a}$

$$\text{Dirichlet}(\alpha_1, \alpha_2, \dots, \alpha_i + 1, \dots, \alpha_S)$$

- After $n_{s,a} = \alpha_1 + \dots + \alpha_S$ observations for a state-action pair s, a
 - Posterior mean vector is empirical mean

$$\hat{P}_{s,a}(i) = \frac{\alpha_i}{\alpha_1 + \dots + \alpha_S} = \frac{\alpha_i}{n_{s,a}}$$

- variance bounded by $\frac{1}{n_{s,a}}$
- With more trials of s, a , the posterior mean concentrates around true mean

Posterior Sampling for RL (Thompson Sampling)

Learning

- Maintain a Dirichlet posterior for $P_{s,a}$ for every s, a
 - After round t , on observing outcome s_{t+1} , update for state s_t and action a_t

To decide action

- Sample a $\tilde{P}_{s,a}$ for every s, a
- Compute the optimal policy $\tilde{\pi}$ for sample MDP $(S, A, \tilde{P}, r, s_0)$
- Choose $a_t = \tilde{\pi}(s_t)$

Exploration-exploitation

- Exploitation: With more observations Dirichlet posterior **concentrates**, $\tilde{P}_{s,a}$ approaches empirical mean $\hat{P}_{s,a}$
- Exploration: **Anti-concentration** of Dirichlet ensures exploration for states/actions/policies less explored

Posterior Sampling [A., Jia, NeurIPS 2017]: general structure

- Proceed in epochs, an epoch ends when the number of visits of *some* state-action pair doubles.

In the beginning of every epoch k

- Update posterior using samples from previous epoch.
- Use **samples from posterior** to set MDP $(S, A, r, \tilde{P}, s_1)$
 - ~~MDP with value greater than true MDP (Upper bound!!)~~
- Solve the **sample MDP** to find optimal policy $\tilde{\pi}$

Execute $\tilde{\pi}$ in epoch k

- observe samples s_t, a_t, r_t, s_{t+1}

Go to next epoch If visits of *some* state-action pair doubles

- If $n_k(s, a) \geq 2 n_{k-1}(s, a)$ for some s, a

Optimistic Posterior Sampling [A., Jia, NeurIPS 2017]

In every epoch

- For every s, a , generate **multiple** $\psi = \tilde{O}(S)$ independent samples from a Dirichlet posterior for $P_{s,a}$
- Form **extended** sample MDP $(S, \psi A, \tilde{P}, r, s_0)$
- Find optimal policy $\tilde{\pi}$ and use through the epoch

Further, initial exploration:

- For s, a with very small $N_{s,a} < \sqrt{\frac{TS}{A}}$, use a simple optimistic sampling, that provides extra exploration

Main result [A., Jia NeurIPS 2017]

- An algorithm **based on posterior sampling** with high probability near-optimal worst-case regret upper bound
- Recall regret:

$$\text{Regret}(M, T) = T \rho^* - \sum_{t=1}^T r(s_t, a_t)$$

- Theorem: For any **communicating** MDP M with (unknown) diameter D , and for $T \geq S^5 A$, with high probability:

$$\text{Regret}(M, T) \leq \tilde{O}(DS\sqrt{AT})$$

Next...

- UCRL
- Posterior sampling algorithm for MDPs
- **Proof techniques**

Regret Analysis (outline)

- Average regret in an epoch k

$$\rho^* - \frac{1}{T_k} \sum_{t \in T_k} r_{s_t, a_t} = (\rho^* - \tilde{\rho}) + \left(\tilde{\rho} - \frac{1}{T_k} \sum_{t \in T_k} r_{s_t, a_t} \right)$$

- Optimal Gain (Asymptotic average reward) for true MDP M (policy π^*)
- Optimal Gain for sample/optimistic MDP (policy $\tilde{\pi}_k$)
- **First term:** we show optimism $\tilde{\rho} \geq \rho^*$
 - Immediate for UCB
 - Needs to use spread (anti-concentration) of posterior in Thompson Sampling

Regret Analysis (main insights)

- Average regret in an epoch k

$$\rho^* - \frac{1}{T_k} \sum_{t \in T_k} r_{s_t, a_t} = \overbrace{(\rho^* - \tilde{\rho})}^{\leq 0} + \left(\tilde{\rho} - \frac{1}{T_k} \sum_{t \in T_k} r_{s_t, a_t} \right)$$

- Optimal Gain for true MDP M (policy π^*)
- Optimal Gain for sample/optimistic MDP (policy $\tilde{\pi}_k$)

- **Second term**

- Same policy $\tilde{\pi}_k$ but on different MDs
- $\tilde{\rho}$: MDP with estimated/sampled transition probability vector
- $\frac{1}{T_k} \sum_{t \in T_k} r_{s_t, a_t}$: True MDP, true transition probability
- Bounded using **concentration** of estimated/sampled transition probability vector

Gain and bias for communicating MDP

- Bellman equations relating optimal average reward to per state reward: $\forall s$,

$$\lambda^* = r_{s,\pi^*(s)} + P_{s,\pi^*(s)} \cdot h^* - h_s^*$$

- $h^* \in R^S$ is called **bias vector** and satisfies $|h_i - h_j| \leq D$ for all $i, j \in S$

Optimism

Two main results

Optimism of transition matrix on a projection is sufficient

$$\tilde{\rho} \geq \rho^* \text{ if for every } s, a, \tilde{P}_{s,a} \cdot h^* \geq P_{s,a} \cdot h^*$$

If a set of samples satisfy optimism on projection to *unknown* bias vector h^*

Anti-concentration of Dirchlet posterior:

For any fixed bounded vector h , a sample satisfies above with probability $1/S$

- improved to a constant in the recent work by Tiapkin et al. 2022
- No need to know h^* !
- (need multiple $O(S \log(\frac{SA}{\rho}))$ samples for high probability)

Concentration

- Recall Lemma: With high probability, sample extended MDP is a communicating MDP with diameter at most $2D$
- Recall Bellman equations for communicating MDP: for all s

$$\tilde{\rho} = r_{s,\pi^*(s)} + \tilde{P}_{s,\pi^*(s)} \cdot \tilde{\mathbf{h}} - \tilde{h}_s$$

- $\tilde{\mathbf{h}} \in R^S$ is the bias vector of sample and satisfies $|\tilde{h}_i - \tilde{h}_j| \leq 2D$ for all $i, j \in S$

$$\left(\tilde{\rho} - \frac{1}{T_k} \sum_{t \in T_k} r_{s_t, a_t} \right) = \frac{1}{T_k} \sum_{t \in T_k} (\tilde{P}_{s_t, a_t} \cdot \tilde{\mathbf{h}} - \tilde{h}_{s_t})$$

- For true model, $P_{s_t, a_t} \cdot \tilde{\mathbf{h}} = \mathbb{E}[\tilde{\mathbf{h}} \cdot \mathbf{1}_{s_{t+1}}] = \mathbb{E}[\tilde{h}_{s_{t+1}}]$

Bounding the second term

$$\left(\tilde{\lambda} - \frac{1}{T_k} \sum_{t \in T_k} r_{s_t, a_t} \right) = \frac{1}{T_k} \sum_{t \in T_k} (\tilde{P}_{s_t, a_t} \cdot \tilde{\mathbf{h}} - P_{s_t, a_t} \cdot \tilde{\mathbf{h}} + P_{s_t, a_t} \cdot \tilde{\mathbf{h}} - \tilde{h}_{s_t})$$

- Bound deviation of posterior sample from true model $(\tilde{P}_{s,a} - P_{s,a}) \cdot \tilde{\mathbf{h}}$
- Challenge: $\tilde{\mathbf{h}}$ is not fixed.
- Union bound over $\tilde{\mathbf{h}}$ gives an extra $\sqrt{S}$.
- Removed for episodic case by Tiapkin et al. 2022.

Optimistic Q-learning [Jin, Allen-Zhu, Bubeck, Jordan 2018]

- Model-free algorithm
 - Learning Q-values and not explicitly the reward/transition model
- Episodic setting
 - Finite horizon H
 - Dynamic programming solution for known MDP
 - Non-stationary policy (different optimal Q-value functions at different steps of the horizon)
 - $\pi_h(s, a) = \arg \max Q_h(s, a)$

Recall Q-learning (stationary)

Initialize $Q(s, a), \forall s, a$

For $t = 1, 2, \dots$,

- Observe current state s
- Take action $a = \arg \max_j Q^t(s, j)$
- Observe reward r , next state s'
- Update

$$Q^{t+1}(s, a) \leftarrow (1 - \alpha_k) Q^t(s, a) + \alpha_k (r_t + \gamma \max_{a'} Q^t(s', a'))$$

(with some added exploration)

Converges to stationary policy $\pi^(s) = \operatorname{argmax}_a Q^*(s, a)$*

Q-learning episodic

Initialize $Q_h(s, a), \forall s, a, h = 1, \dots, H$

For episodes $k = 1, 2, \dots$

For $h = 1, \dots, H$

- Observe state s
- Take action $a = \arg \max_j Q_h^k(s, j)$
- Observe reward r , next state s'
- Update

$$Q_h^{k+1}(s, a) \leftarrow (1 - \alpha_k) Q_h^k(s, a) + \alpha_k (r + \gamma \max_{a'} Q_{h+1}^k(s', a'))$$

(with some added exploration)

Converges to non-stationary policy $\pi_h^(s) = \operatorname{argmax}_a Q_h^*(s, a)$*

Optimistic Q-learning episodic [Jin et al. 2018]

Initialize $Q_{\textcolor{red}{h}}(s, a), \forall s, a, h = 1, \dots, H$

For episodes $k = 1, 2, \dots$

For $h = 1, \dots, H$

- Observe state s
- Take action $a = \arg \max_j Q_{\textcolor{red}{h}}^k(s, j)$
- Observe reward r , next state s'
- Update

$$b_k(s, a) := \tilde{O}\left(\frac{H^{3/2}}{\sqrt{N_k(s, a)}}\right)$$
$$\alpha_k = \frac{H + 1}{H + N_k(s, a)}$$

$$Q_{\textcolor{red}{h}}^{k+1}(s, a) \leftarrow (1 - \alpha_k) Q_{\textcolor{red}{h}}^k(s, a) + \alpha_k \left(r + \gamma \max_{a'} Q_{\textcolor{red}{h}+1}^k(s', a') \right) + \textcolor{red}{b}_k(s, a)$$

Regret bound [Jin et al. 2018]

- $\tilde{O}(\sqrt{H^3 SAT})$ worst-case regret bound
- Main challenge compared to model-free algorithms:
 - For any given s, a , model based algorithms can update the models $\tilde{R}(s, a), \tilde{P}(s, a)$ using unbiased samples from the reward and transition model
 - Q-values are updated in a biased way
 - With more samples, convergence to the true Q^* -values does not follow directly from concentration

takes action a^j , where $a = \pi^*(s)$, and j is a sample satisfying above inequality. Note that π is essentially π^* but with a different transition probability model. Let Q_π be the transition matrix for this policy, whose rows are formed by the vectors $Q_{s, \pi^*(s)}^{j,k}$, and P_{π^*} be the transition matrix whose rows are formed by the vectors $P_{s, \pi^*(s)}$. Above implies

$$Q_\pi h^* \geq P_{\pi^*} h^* - \delta \mathbf{1}.$$

Let Q_π^* denote the limiting matrix for Markov chain with transition matrix Q_π . Observe that Q_π is aperiodic, recurrent and irreducible : it is aperiodic and irreducible because each entry of Q_π being a sample from Dirichlet distribution is non-zero, and it is positive recurrent because in a finite irreducible Markov chain, all states are positive and recurrent. This implies that Q_π^* is of the form $\mathbf{1} \mathbf{q}^{*T}$ where $\mathbf{q}^*$ is the stationary distribution of Q_π , and $\mathbf{1}$ is the vector of all 1s (refer to (A.6) in Puterman [2014]). Also, $Q_\pi^* Q_\pi = Q_\pi$, and $Q_\pi^* \mathbf{1} = \mathbf{1}$.

Backup

Therefore, the gain of policy π

$$\tilde{\lambda}(\pi)\mathbf{1} = (r_{\pi}^T \mathbf{q}^*)\mathbf{1} = Q_{\pi}^* r_{\pi}$$

where r_{π} is the S dimensional vector $[r_{s,\pi(s)}]_{s=1,\dots,S}$. Now,

$$\begin{aligned}\tilde{\lambda}(\pi)\mathbf{1} - \lambda^*\mathbf{1} &= Q_{\pi}^* r_{\pi} - \lambda^*\mathbf{1} \\ &= Q_{\pi}^* r_{\pi} - \lambda^*(Q_{\pi}^* \mathbf{1}) \quad \dots (\text{using } Q_{\pi}^* \mathbf{1} = \mathbf{1}) \\ &= Q_{\pi}^* (r_{\pi} - \lambda^*\mathbf{1}) \\ &= Q_{\pi}^* (I - P_{\pi^*})h^* \quad \dots (\text{using (1)}) \\ &= Q_{\pi}^* (Q_{\pi} - P_{\pi^*})h^* \quad \dots (\text{using } Q_{\pi}^* Q_{\pi} = Q_{\pi}^*) \\ &\geq -\delta\mathbf{1} \quad \dots (\text{using } (Q_{\pi} - P_{\pi^*})h^* \geq -\delta\mathbf{1}, Q_{\pi}^* \mathbf{1} = \mathbf{1}).\end{aligned}$$

Then, by optimality,

$$\tilde{\lambda}_k \geq \tilde{\lambda}(\pi) \geq \lambda^* - \delta.$$

PAC analysis: learning from experience

- Bound number of steps on which suboptimal policy is played
- Play near-optimal on all but a polynomial number of steps (Kakade, 2003; Strehl and Littman, 2008b, Strehl et al 2006, 2009)
- Example: [Strehl et al 2006] (has been improved in more recent work)

Theorem 1 *Let M be any MDP and let ϵ and δ be two positive real numbers. If Delayed Q-learning is executed on MDP M , then it will follow an ϵ -optimal policy on all but $O\left(\frac{SA}{(1-\gamma)^8 \epsilon^4} \ln \frac{1}{\delta} \ln \frac{1}{\epsilon(1-\gamma)} \ln \frac{SA}{\delta \epsilon(1-\gamma)}\right)$ timesteps, with probability at least $1 - \delta$.*

Generative model: Sample complexity for ϵ optimal

Some examples:

- An $O(SA)$ analysis of Q-learning.
 - Michael Kearns and Satinder Singh, Finite-Sample Convergence Rates for Q-learning and Indirect Algorithms NIPS 1999.
 - Sheng Zhang, Zhe Zhang, Siva Theja Maguluri, Finite Sample Analysis of Average-Reward TD Learning and Q-Learning, NeurIPS 2021
- An analysis of TD-lambda.
 - Michael Kearns and Satinder Singh, Bias-Variance Error Bounds for Temporal Difference Updates, COLT 2000.
- Analysis of Policy gradient methods
 - Approximately optimal Approximate RL, Kakade and Langford 2002
 - On the Theory of Policy Gradient Methods: Optimality, Approximation, and Distribution Shift, Alekh Agarwal, Sham M. Kakade, Jason D. Lee, Gaurav Mahajan, 2020

Learning complexity analysis

- Optimal learning algorithm?
 - “optimally explore” meaning to obtain maximum expected discounted reward $E[\sum_{t=1}^{\infty} \gamma^{t-1} r_t]$ over *a known prior* over the unknown MDPs
 - Tractable in special cases ($\gamma < 1$, *MAB*, Gittins, 1989).

Regret analysis

- Bound difference in total reward of algorithm compared to a benchmark policy
 - $Reg(M, T) = T \rho^*(s_0) - E[\sum_{t=1}^T r_t]$
 - $\rho^*(s_0)$ is infinite horizon average reward achieved by the best stationary policy π^*
- Episodic/single thread
 - Episodic: starting state is reset every H steps
 - Single thread: assume Communicating with diameter D
- Bayesian regret bounds: (expectation over known prior on MDP M); example
 - Theorem [Osband, Russo, Van Roy, 2013]: For **any known prior distribution** f on MDP M, our algorithm achieves in episodic setting: $E_{M \sim f}[Regret(M, T)] \leq \tilde{O}(HS\sqrt{AT})$
- Worst-case regret bounds; example
 - Theorem [Agrawal, Jia, 2017]: For **any communicating MDP** M with (unknown) diameter D, and for $T \geq S^5 A$, our algorithm achieves: $Regret(M, T) \leq \tilde{O}(DS\sqrt{AT})$